Sre Cibi Varsshan K

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EDUCATION

Vellore Institute of Technology, Chennai | Expected 2027
B.Tech Computer Science and Engineering (Artificial Intelligence and Machine Learning)
Relevant coursework: Computer Graphics, Data Structures and Algorithms, Machine Learning.

SUMMARY

Gameplay and graphics programmer: Unity and C# gameplay systems with playable WebGL builds, a URP post-processing shader in HLSL, real-time rendering in OpenGL and GLSL, and reinforcement-learning agents that play games.

PROJECTS

Doofus Diary - 3D platform-survival game | Unity, C#, Rigidbody physics, WebGL | Aug 2026
Play: xerfzy.github.io/HW_2026_Test/Build | Source: github.com/xerfzy/HW_2026_Test

- Built a gameplay prototype for the Hitwicket game-developer challenge: timed platforms that spawn ahead of the player, fade in and out with coroutines, and shorten their lifetime as the score grows.
- Implemented Rigidbody movement with camera-relative input, Animator-driven states and score-gated single and double jumps; wrote a spring-damper follow camera (explicit stiffness and damping integration) for smooth, momentum-based tracking.
- Structured the game into focused systems (GameManager, ScoreManager, AudioManager, PulpitSpawner) with Normal and Endless modes, world-space countdown UI, and gameplay values loaded from a JSON config.

Slot Machine | Unity 6, C#, URP, HLSL, ShaderLab, WebGL | Apr - May 2026
Play: xerfzy.github.io/Slot_Machine/Build | Source: github.com/xerfzy/Slot_Machine

- Built a three-reel slot machine as an assignment: outcomes chosen up front by weighted random selection over ScriptableObject symbol data, then revealed by coroutine-driven reels with staggered start and stop and pixel-snapped alignment.
- Wrote a URP full-screen post-processing shader in HLSL combining palette quantisation, 8x8 Bayer ordered dithering and Sobel edge detection; added betting UI, audio and a scripted croupier commentary system.

SDF Ray Marcher | C++, OpenGL, GLSL | Jan - Apr 2026
github.com/xerfzy/Ray_marcher_using_fragment_shader_in_opengl

- Built a real-time ray marcher in a GLSL fragment shader (signed distance functions, domain repetition, central-difference normals) with an FPS-style free-fly camera and a framebuffer-object Sobel post-processing pass.

Space Shooter DQN - game AI (coursework) | Python, PyTorch, Pygame | Apr 2026
github.com/xerfzy/Space_shooter_using_DQN_vc

- Built a Pygame shooter with waves of enemies on sine, straight and zigzag patterns, and trained a Dueling Double DQN that plays from four stacked 84x84 grayscale frames using a 50,000-transition replay buffer and Huber loss.

TrackMania TD3 - driving agent research | Python, PyTorch, TMRL | Sep 2026
github.com/thenamelessone888/RL

- Implemented TD3 from scratch for TrackMania 2020 with LIDAR observations and a warm start cloned from recorded human driving; diagnosed a recurring wall collision from evaluation traces and added a stuck-recovery reward penalty.

RESEARCH

Indian Patent Application IN202641060337 A1 (published application, not granted) | May 2026
"A System for Swarm of Drone for Crowed Behaviour Analysis" (title as published). Application No. 202641060337, filed 12 May 2026, published 22 May 2026. Inventors: Sre Cibi Varsshan K, D Kavitha. Applicant: Vellore Institute of Technology, Chennai.

TECHNICAL SKILLS

Game development: Unity 6, C#, URP, ShaderLab, HLSL, Rigidbody physics, Animator, coroutines, ScriptableObjects, Unity UI, WebGL builds
Graphics: OpenGL, GLSL fragment and compute shaders, framebuffer objects, ray marching, ray tracing
Game AI: Deep Q-Networks, TD3, behaviour cloning, PyTorch, Pygame
Languages and tools: C#, C++, Python, Git, Linux